COURSE 6 · BIOLOGY

Biology

Printable workbook with worked solutions

72 problems across 9 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 72 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook, which checks answers instantly and adds a quick quiz for each module.

> **A good routine for each module**
>
> Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then redo anything you got wrong without looking.

Module 6.1

The chemistry of life

6.1.1 ●○○

Proteins are polymers of

A. sugars

B. amino acids

C. fatty acids

D. nucleotides

6.1.2 ●○○

Which molecule forms cell membranes?

A. starch

B. phospholipid

C. DNA

D. glucose

6.1.3 ●●○

Why does an enzyme stop working at high temperature?

A. it is used up

B. its shape changes and the active site no longer fits

C. it becomes a carbohydrate

D. substrates evaporate

6.1.4 ●●○

Why does a lake freeze from the top down, protecting fish?

A. ice is denser than water

B. ice is less dense than water and floats

C. the bottom is warmer from the core

D. fish produce heat

6.1.5 ●●○

Using v = Vmax[S]/(Km + [S]), find v when [S] = Km and Vmax = 10 units.

Answer: _____ units

6.1.6 ●●○

Breaking starch into glucose during digestion is

A. condensation

B. hydrolysis

C. oxidation

D. fusion

6.1.7 ●●○

At very high substrate concentration, enzyme rate levels off because

A. the enzyme denatures

B. all active sites are occupied

C. products block it

D. substrate runs out

6.1.8 ●●●

Explain how an enzyme's specificity follows from its structure.

Module 6.2

Cells

6.2.1 ●○○

Which cells lack a nucleus?

A. plant cells

B. animal cells

C. bacteria

D. fungal cells

6.2.2 ●○○

Where in the cell is aerobic respiration carried out?

A. nucleus

B. mitochondrion

C. ribosome

D. cell wall

6.2.3 ●●○

A cell 20 µm across appears 8 mm across in a micrograph. What is the magnification?

Answer: _____ ×

6.2.4 ●●○

What is the surface-area-to-volume ratio of a cube-shaped cell with sides 2 µm? (μm^{-1})

Answer: _____

6.2.5 ●●○

And for sides of 10 µm?

Answer: _____

6.2.6 ●●○

A red blood cell is placed in pure water. What happens?

A. it shrinks

B. water enters by osmosis and it swells, possibly bursting

C. nothing

D. it divides

6.2.7 ●●○

Which process needs energy from the cell?

A. diffusion

B. osmosis

C. active transport

D. all of these

6.2.8 ●●●

What evidence suggests mitochondria were once free-living bacteria?

Module 6.3

Metabolism

6.3.1 ●○○

Which gas is released by photosynthesis?

A. carbon dioxide

B. oxygen

C. nitrogen

D. methane

6.3.2 ●○○

What is the cell's main energy currency?

A. glucose

B. ATP

C. DNA

D. starch

6.3.3 ●●○

How many molecules of O_2 are needed to respire one molecule of glucose?

Answer: _____

6.3.4 ●●○

Glucose has M = 180 g/mol. How many moles of CO_2 are produced when 90 g of glucose is fully respired?

Answer: _____ mol

6.3.5 ●●○

Aerobic respiration gives about 30 ATP per glucose; anaerobic gives 2. How many times more ATP does aerobic give?

Answer: _____ ×

6.3.6 ●●○

During a sprint, muscle cells short of oxygen produce

A. ethanol

B. lactate

C. glucose

D. oxygen

6.3.7 ●●○

Why do leaves look green?

A. chlorophyll absorbs green light

B. chlorophyll reflects green light while absorbing red and blue

C. they contain green sugar

D. water is green

6.3.8 ●●●

Explain why oxygen is essential for producing most of a cell's ATP.

Module 6.4

DNA, genes and protein synthesis

6.4.1 ●○○

Which base pairs with guanine in DNA?

A. adenine

B. thymine

C. cytosine

D. uracil

6.4.2 ●○○

Write the complementary strand to 5′-ATGCCT-3′ (reading the partner 3′ to 5′).

A. TACGGA

B. ATGCCT

C. UACGGA

D. GCATTC

6.4.3 ●●○

A DNA sample is 30% adenine. What percentage is guanine?

Answer: _____ %

6.4.4 ●●○

How many amino acids are coded by an mRNA sequence of 300 bases (ignoring stop codons)?

Answer: _____

6.4.5 ●●○

In transcription, thymine in DNA is matched in RNA by

A. adenine

B. uracil

C. thymine

D. cytosine

6.4.6 ●●○

Why can a mutation have no effect on the protein?

A. mutations never change DNA

B. the new codon may code for the same amino acid

C. proteins ignore DNA

D. ribosomes fix all errors

6.4.7 ●●○

Semi-conservative replication means each new DNA molecule has

A. two new strands

B. two old strands

C. one old and one new strand

D. random mixtures

6.4.8 ●●●

Why is the near-universal genetic code evidence for common ancestry?

Module 6.5

Inheritance and genetics

6.5.1 ●○○

Two Aa parents. What fraction of offspring are expected to be aa? Give a decimal.

Answer: _____

6.5.2 ●○○

AA × aa. What fraction of offspring are heterozygous? Give a decimal.

Answer: _____

6.5.3 ●●○

Aa × aa. What fraction show the dominant phenotype? Give a decimal.

Answer: _____

6.5.4 ●●○

Two carriers of cystic fibrosis (recessive) have three children. What is the probability none are affected? Give a decimal.

Answer: _____

6.5.5 ●●○

A woman who is a carrier of colour blindness (X^BX^b) has children with a man with normal vision. What fraction of their sons are expected to be colour blind?

A. 0

B. 1/4

C. 1/2

D. all

6.5.6 ●●○

A recessive allele has frequency q = 0.1. Using Hardy–Weinberg, what fraction of the population is homozygous recessive? Give a decimal.

Answer: _____

6.5.7 ●●●

Using Hardy–Weinberg with q = 0.1, what fraction of the population are carriers (heterozygous)? Give a decimal.

Answer: _____

6.5.8 ●●●

Why do siblings with the same parents look different?

Module 6.6

Evolution and natural selection

6.6.1 ●○○

Which three things are needed for natural selection?

A. mutation, migration, mating

B. variation, inheritance, differential reproduction

C. need, effort, change

D. size, speed, strength

6.6.2 ●○○

Antibiotic resistance in bacteria is an example of

A. Lamarckism

B. natural selection

C. genetic drift only

D. spontaneous generation

6.6.3 ●●○

An allele has frequency 0.5 and a 10% advantage ($s = 0.1$). Using $p' = p(1+s)/[p(1+s) + (1 - p)]$, what is its frequency next generation?

Answer: _____

6.6.4 ●●○

A human arm, a bat's wing and a whale's flipper share the same bone pattern. These are

A. analogous structures

B. homologous structures

C. vestigial structures

D. coincidences

6.6.5 ●●○

Genetic drift has the biggest effect in

A. large populations

B. small populations

C. all populations equally

D. no populations

6.6.6 ●●○

A giraffe stretching its neck to reach leaves will

A. pass on a longer neck

B. not change its offspring's neck length

C. become a new species

D. evolve during its life

6.6.7 ●●○

What usually starts the formation of a new species?

A. a single mutation

B. separation of populations so they diverge

C. a change in diet

D. climate warming alone

6.6.8 ●●●

Explain why finishing a course of antibiotics matters, using natural selection.

Module 6.7

Human body systems

6.7.1 ●○○

A resting heart beats 70 times a minute with a stroke volume of 70 mL. What is the cardiac output in litres per minute?

Answer: _____ L/min

6.7.2 ●○○

Where in the lungs does gas exchange occur?

A. trachea

B. bronchi

C. alveoli

D. diaphragm

6.7.3 ●●○

Which hormone lowers blood glucose?

A. glucagon

B. adrenaline

C. insulin

D. thyroxine

6.7.4 ●●○

Sweating when hot is an example of

A. positive feedback

B. negative feedback

C. a reflex arc only

D. digestion

6.7.5 ●●○

During exercise, heart rate rises to 150 beats/min and stroke volume to 110 mL. What is cardiac output, in L/min?

Answer: _____ L/min

6.7.6 ●●○

A nerve impulse travels at 80 m/s. How long does it take to travel 1.6 m from toe to brain, in milliseconds?

Answer: _____ ms

6.7.7 ●●○

Compared with nervous signals, hormonal signals are generally

A. faster and briefer

B. slower and longer-lasting

C. electrical

D. only local

6.7.8 ●●●

Explain how the body corrects a fall in core temperature.

Module 6.8

Microbes, immunity and disease

6.8.1 ●○○

A bacterium divides every 20 minutes. Starting from one cell, how many are there after 2 hours?

Answer: _____

6.8.2 ●○○

Antibiotics are effective against

A. viruses

B. bacteria

C. both

D. neither

6.8.3 ●●○

Measles has $R_0 \approx 15$. What is the herd immunity threshold, as a percentage?

Answer: _____ %

6.8.4 ●●○

A disease has $R_0 = 3$. What percentage must be immune for herd immunity?

Answer: _____ %

6.8.5 ●●○

With $R_0 = 4$ and 60% of people immune, what is R_eff?

Answer: _____

6.8.6 ●●○

Vaccines work by

A. killing all microbes in the body

B. creating immune memory without causing the disease

C. acting as antibiotics

D. replacing white blood cells

6.8.7 ●●○

Antibodies are made by

A. red blood cells

B. B cells

C. platelets

D. neurons

6.8.8 ●●●

Why can a person who is not vaccinated still be protected by others' vaccination?

Module 6.9

Ecology and ecosystems

6.9.1 ●○○

Producers in an ecosystem capture 20 000 kJ. About how much energy reaches the carnivores that eat herbivores?

Answer: _____ kJ

6.9.2 ●○○

Which process removes CO_2 from the atmosphere?

A. respiration

B. combustion

C. photosynthesis

D. decomposition

6.9.3 ●●○

Why are there fewer top predators than herbivores?

A. predators breed slowly only

B. energy is lost at each trophic level

C. predators are larger

D. herbivores hide

6.9.4 ●●○

In the logistic model, what is dN/dt when $N = K/2$, $r = 0.4$ per year and $K = 1000$?

Answer: _____ per year

6.9.5 ●●○

And what is dN/dt when N = K?

Answer: _____ per year

6.9.6 ●●○

Atmospheric CO_2 rose from 280 ppm to 420 ppm. What percentage increase is that?

Answer: _____ %

6.9.7 ●●○

In predator–prey cycles, predator population peaks typically

A. come before prey peaks

B. follow prey peaks

C. coincide with them

D. never occur

6.9.8 ●●●

Explain why eating lower on the food chain can feed more people from the same land.

Capstone

Tracking an outbreak

A new bacterial infection appears in a town of 20 000 people. Health officials use biology from this course to understand and control it. Work through their analysis.

Part 1 ●●○

In the lab, the bacterium doubles every 30 minutes. Starting from 100 cells, how many are there after 4 hours?

Answer: _____

Part 2 ●●○

Each case infects 2.5 others in a susceptible population ($R_0 = 2.5$). What vaccination coverage gives herd immunity, as a percentage?

Answer: _____ %

Part 3 ●●○

If 50% of the town is immune, what is R_eff?

Answer: _____

Part 4 ●●○

A patient asks for antibiotics for a cold during the outbreak. The best response is

A. prescribe them as a precaution

B. explain that colds are viral and antibiotics won't help, and overuse breeds resistance

C. give a double dose

D. use antiviral antibiotics

Part 5 ●●○

The bacterium's toxin gene has a 300-base coding sequence. How many amino acids long is the toxin protein (ignoring the stop codon)?

Answer: _____

Part 6 ●●●

Some isolates are now resistant to the first-choice antibiotic. Explain how this happened and suggest two ways to slow it.

Answers

Worked solutions

Module 6.1 · The chemistry of life

6.1.1 ANSWER B. amino acids

1 Twenty kinds of amino acid.

6.1.2 ANSWER B. phospholipid

1 Lipids with a water-loving head and water-avoiding tails.

6.1.3 ANSWER B. its shape changes and the active site no longer fits

1 Denaturation.

6.1.4 ANSWER B. ice is less dense than water and floats

1 Hydrogen bonding gives ice an open structure.

6.1.5 ANSWER 5 units

1 $v = 10 \times Km/(2Km) = 5$: half the maximum.

6.1.6 ANSWER B. hydrolysis

1 Water is used to split the bonds.

6.1.7 ANSWER B. all active sites are occupied

1 Saturation.

6.1.8 ANSWER An enzyme is a protein chain folded into a precise three-dimensional shape. The active site is a pocket whose shape and pattern of charges match particular substrate molecules. Only substrates that fit and interact correctly can bind and react, so each enzyme catalyses one reaction or a small family of reactions.

1 Think about shape and the active site.

Module 6.2 · Cells

6.2.1 ANSWER C. bacteria

1 Prokaryotes have no nucleus.

6.2.2 ANSWER B. mitochondrion

1 The powerhouse.

6.2.3 ANSWER 400 ×

1 8 mm = 8000 µm.

2 8000 ÷ 20 = 400.

6.2.4 ANSWER 3

1 $6/L = 6/2 = 3\ \mu m^{-1}$.

6.2.5 ANSWER 0.6

1 $6/10 = 0.6\ \mu m^{-1}$: five times smaller.

6.2.6 ANSWER B. water enters by osmosis and it swells, possibly bursting

1 Water moves towards the higher solute concentration inside the cell.

6.2.7 ANSWER C. active transport

1 Moving substances against a gradient.

6.2.8 ANSWER Mitochondria have their own circular DNA, like bacteria; they have their own bacteria-like ribosomes; they divide by splitting, as bacteria do; and they are surrounded by a double membrane, consistent with being engulfed. Gene sequences show their closest relatives are a group of bacteria.

1 Compare mitochondria with bacteria.

Module 6.3 · Metabolism

6.3.1 ANSWER B. oxygen

1 Water is split, releasing O_2.

6.3.2 ANSWER B. ATP

1 Adenosine triphosphate.

6.3.3 ANSWER 6

1 From the balanced equation.

6.3.4 ANSWER 3 mol

1 90 g = 0.5 mol glucose.

2 × 6 = 3 mol CO_2.

6.3.5 ANSWER 15 ×

1 30 ÷ 2.

6.3.6 ANSWER B. lactate

1 Anaerobic respiration in animals.

6.3.7 ANSWER B. chlorophyll reflects green light while absorbing red and blue

1 Green is the colour least absorbed.

6.3.8 ANSWER In aerobic respiration, electrons from glucose pass along the electron transport chain in the mitochondria, releasing energy used to make most of the ATP. Oxygen is the final electron acceptor at the end of the chain, combining with electrons and hydrogen to form water. Without oxygen the chain backs up and stops, leaving only the 2 ATP from glycolysis.

1 What is oxygen's role in the electron transport chain?

Module 6.4 · DNA, genes and protein synthesis

6.4.1 ANSWER C. cytosine

1 G–C.

6.4.2 ANSWER A. TACGGA

1 A→T, T→A, G→C, C→G.

6.4.3 ANSWER 20 %

1 A = T = 30%, so A + T = 60%.

2 G + C = 40%, G = 20%.

6.4.4 ANSWER 100

1 3 bases per codon: 300 ÷ 3.

6.4.5 ANSWER A. adenine

1 The RNA base opposite a DNA T is A. (RNA uses U in place of T, opposite DNA's A.)

6.4.6 ANSWER B. the new codon may code for the same amino acid

1 The code is redundant.

6.4.7 ANSWER C. one old and one new strand

1 Confirmed by Meselson and Stahl in 1958.

6.4.8 ANSWER The assignment of codons to amino acids is largely arbitrary chemically; many other codes would work. Yet bacteria, plants, fungi and animals almost all use the same one. The simplest explanation is that all life inherited it from a common ancestor, just as shared quirks in languages point to a common origin.

1 Is there a chemical reason the code must be this one?

Module 6.5 · Inheritance and genetics

6.5.1 ANSWER 0.25

1 One square in four.

6.5.2 ANSWER 1

1 Every offspring gets A from one parent and a from the other.

6.5.3 ANSWER 0.5

1 Offspring: Aa, Aa, aa, aa.

6.5.4 ANSWER 0.4219

1 Each child: 3/4 unaffected.

2 $(3/4)^3 \approx 0.42$.

6.5.5 ANSWER C. 1/2

1 Sons get Y from father and one X from mother: half get X^b.

6.5.6 ANSWER 0.01

1 $q^2 = 0.01$.

6.5.7 ANSWER 0.18

1 $2pq = 2 \times 0.9 \times 0.1 = 0.18$.

6.5.8 ANSWER Each parent's sex cells are made by meiosis. Independent assortment of 23 chromosome pairs gives over 8 million combinations, and crossing over shuffles alleles within chromosomes. Each child comes from a different egg and sperm, so siblings (except identical twins) inherit different combinations of their parents' alleles.

1 What does meiosis do to allele combinations?

Module 6.6 · Evolution and natural selection

6.6.1 ANSWER B. variation, inheritance, differential reproduction

1 Darwin's core logic.

6.6.2 ANSWER B. natural selection

1 Resistant variants survive and reproduce.

6.6.3 ANSWER 0.5238

1 $p(1 + s) = 0.55$.

2 $0.55 \div (0.55 + 0.5) \approx 0.524$.

6.6.4 ANSWER B. homologous structures

1 Same structure from a common ancestor.

6.6.5 ANSWER B. small populations

1 Chance fluctuations matter more when numbers are small.

6.6.6 ANSWER B. not change its offspring's neck length

1 Acquired characteristics are not inherited.

6.6.7 ANSWER B. separation of populations so they diverge

1 Isolation then divergence.

6.6.8 ANSWER A bacterial population varies in susceptibility. Antibiotics kill the most susceptible first. Stopping early leaves the hardiest survivors, which multiply and pass on their resistance genes, so the next infection is harder to treat. A full course aims to kill the whole population before resistant variants take over.

1 Which bacteria survive a short course?

Module 6.7 · Human body systems

6.7.1 ANSWER 4.9 L/min

1 70 × 70 = 4900 mL/min.

6.7.2 ANSWER C. alveoli

1 Thin-walled air sacs.

6.7.3 ANSWER C. insulin

1 Insulin, from the pancreas.

6.7.4 ANSWER B. negative feedback

1 The response counters the change.

6.7.5 ANSWER 16.5 L/min

1 150 × 110 = 16 500 mL/min.

6.7.6 ANSWER 20 ms

1 t = 1.6/80 = 0.02 s.

6.7.7 ANSWER B. slower and longer-lasting

1 They travel in the blood.

6.7.8 ANSWER Temperature receptors detect the drop and the hypothalamus triggers responses: blood vessels in the skin narrow (vasoconstriction) so less heat is lost; muscles shiver, releasing heat from respiration; hair erects slightly. As temperature returns to normal, the responses switch off: negative feedback.

1 Detect, respond, switch off.

Module 6.8 · Microbes, immunity and disease

6.8.1 ANSWER 64

1 2 hours = 6 doublings.

2 $2^6 = 64$.

6.8.2 ANSWER B. bacteria

1 They target bacterial structures.

6.8.3 ANSWER 93.33 %

1 $1 - 1/15 \approx 0.933$.

2 About 93%.

6.8.4 ANSWER 66.67 %

1 $1 - 1/3 \approx 67\%$.

6.8.5 ANSWER 1.6

1 Susceptible fraction 0.4.

2 $4 \times 0.4 = 1.6$: still spreading.

6.8.6 ANSWER B. creating immune memory without causing the disease

1 They train the adaptive immune system.

6.8.7 ANSWER B. B cells

1 B lymphocytes.

6.8.8 ANSWER If enough of the population is immune, each infected person meets too few susceptible people to pass the infection on to more than one other on average (R_eff below 1). Chains of transmission die out before reaching the unvaccinated, who are shielded indirectly. This is herd immunity.

1 Think about R_eff.

Module 6.9 · Ecology and ecosystems

6.9.1 ANSWER 200 kJ

1 Two steps at 10%: 20 000 → 2000 → 200.

6.9.2 ANSWER C. photosynthesis

1 Plants fix carbon.

6.9.3 ANSWER B. energy is lost at each trophic level

1 Only about 10% passes up each level.

6.9.4 ANSWER 100 per year

1 $dN/dt = rN(1 - N/K) = 0.4 \times 500 \times 0.5$.

2 = 100 per year (the fastest growth).

6.9.5 ANSWER 0 per year

1 $1 - N/K = 0$, so growth stops.

6.9.6 ANSWER 50 %

1 $(420 - 280)/280 = 0.5$.

6.9.7 ANSWER B. follow prey peaks

1 Predators respond to food supply with a lag.

6.9.8 ANSWER Only about 10% of the energy in plants ends up in the animals that eat them. Feeding crops directly to people uses that energy at the first step, while feeding them to livestock loses about 90% before it reaches people as meat. So the same crop area supports roughly ten times as many people eating plants as eating meat from those plants.

1 Apply the 10% rule.

Capstone · Tracking an outbreak

Part 1 ANSWER 2.56e+04

1 8 doublings.

2 $100 \times 2^8 = 25\,600$.

Part 2 ANSWER 60 %

1 $1 - 1/2.5 = 0.6$.

Part 3 ANSWER 1.25

1 $2.5 \times 0.5 = 1.25$: still above 1.

Part 4 ANSWER B. explain that colds are viral and antibiotics won't help, and overuse breeds resistance

1 Antibiotics don't act on viruses.

Part 5 ANSWER 100

1 300 ÷ 3.

Part 6 ANSWER Within the bacterial population there was variation, including a few cells with resistance genes (from mutation or gene transfer). Antibiotic use killed susceptible cells and left resistant ones to multiply, so resistance became common: natural selection. To slow it: prescribe antibiotics only when needed and complete full courses; use infection control (handwashing, isolation) so fewer infections need treatment; rotate or combine drugs; monitor resistance.

1 Variation, selection, inheritance.